

The Aerospace Launch Probability Simulation

Grant Cates
The Aerospace Corporation
15835 Phillips Parkway
Eastern Range, FL 32925
321-867-7820
grant.r.cates@aero.org

Kara Schmitt
The Aerospace Corporation
Greens II 1031
Chantilly, VA 20910
571-307-3707
kara.a.schmitt@aero.org

Abstract— Launch timeliness is an important consideration for mission modeling and optimization. This is especially true for planetary missions and may well be of critical importance for future crew rescue missions. The Aerospace Launch Probability Simulation (ALPS) is a discrete event simulation based tool that produces a cumulative probability distribution estimate for when a space launch is likely to occur relative to the planned launch date at the start of the launch countdown. ALPS considers all factors that have the potential to scrub a launch attempt including problems with the launch vehicle, the payload, which could be a crewed space capsule, and supporting launch range infrastructure as well as the weather constraints. The model can be run with all scrub factors active or with any combination of factors enabled or disabled. Historical data from all Space Shuttle, Atlas V, Delta IV and Falcon 9 launch attempts was analyzed to develop estimates for the scrub factors in terms of their probability of occurrence and the duration of the associated scrub. ALPS can analyze Atlas V, Delta IV, and Falcon 9 launch probabilities in its current state of development. This paper describes all of the scrub risk factors, their basis of estimate, and presents comparative cumulative probability distributions for Atlas V, Delta IV, and Falcon 9. ALPS can be used to ensure mission success when launch timeliness is critical.

TABLE OF CONTENTS

1. INTRODUCTION	1
2. SCRUB FACTORS AND BASIS OF ESTIMATE	
OVERVIEW	2
3. GENERIC SCRUB FACTORS	2
4. LAUNCH VEHICLE SCRUB FACTORS	7
5. MISSION SPECIFIC FACTORS	8
6. CREW SPECIFIC FACTORS	8
7. DESCRIPTION OF ALPS	10
8. RESULTS	11
9. CONCLUSIONS AND FORWARD WORK	14
APPENDIX: LAUNCH VEHICLE SCRUB AND T-0	
DELAY HISTORY	15
ACKNOWLEDGEMENTS	19
REFERENCES	19
BIOGRAPHY	20

1. INTRODUCTION

Once a launch countdown has begun, what is the probability that launch will actually occur during the planned launch window? If the launch attempt for that day is scrubbed, then what is the cumulative probability that launch will occur on

978-1-4673-7676-1/16/\$31.00 ©2016 IEEE

subsequent days? How many days does it take to get to a high cumulative probability of launch? These are the types of questions that the Aerospace Launch Probability Simulation (ALPS) addresses.

Having a high launch probability can be very important but most space missions can tolerate launch delays. This is good because launch countdown delays are quite common. However, when it comes to planetary missions the need to have a high cumulative probability of launch is paramount. Missing the launch opportunity for a mission to Mars will mean a delay of 26 months while waiting for Earth and Mars to come back into alignment. A trans-Mars-injection launch opportunity may last for only a few weeks, but the actual duration is dependent upon many factors. NASA's Curiosity rover had a three-week launch opportunity. The opportunity for NASA's InSight mission extends from March 4 to March 30, 2016.

This paper and ALPS does not address delays leading up to the start of the launch countdown. Such delays can be at least somewhat mitigated by adding margin to the spacecraft and launch vehicle preparations campaign. However, the actual launch cannot be attempted until the opening of the planetary opportunity. Consequently, the cumulative probability of launching during the opportunity is a function of the likelihood of launching on any given attempt beginning with the initial attempt, the time required to make subsequent attempts, and the duration of the planetary opportunity. A cumulative probability distribution for when launch occurs relative to the initial launch attempt becomes a critical measurement. The Aerospace Launch Probability Simulation was created to provide this information.

The launch countdown period can be thought of as beginning at the start of a defined procedure developed by the launch service provider working in partnership with the payload customer and launch range officials. The launch probability simulation can be tailored to begin at that point or alternative points such as launch minus 1 day or launch minus 2 days, etc. Another option is to begin the simulation with the start of propellant loading, which would typically be a few hours prior to launch. Or conversely, the simulation could be started at the beginning of a wet dress rehearsal or static fire test which typically occurs a few days before launch for Falcon 9. Once the launch countdown period has begun in the simulation any factor that results in the launch not occurring on the planned launch day is categorized as a launch scrub.

Section 2 introduces all of the scrub factors considered and modeled while Sections 3 through 6 provide additional details and the basis of estimate for each scrub factor. Section 7 describes ALPS and how it is used to perform the quantitative analysis. Section 8 presents simulation results for Atlas V, Delta IV and Falcon 9 and compares those results to the empirical data. Conclusions and forward work are discussed in Section 9.

2. SCRUB FACTORS AND BASIS OF ESTIMATE OVERVIEW

Launch scrubs occur for a myriad of reasons. In general there are three categories of scrub factors, (1) those that are generic factors that every launch attempt is subject to; (2) launch vehicle specific scrub factors; and (3) payload or mission specific factors. Nineteen unique scrub factors were identified as shown in Table 1, some of which have monthly variability. For example, launch weather falls in the generic category and is subject to monthly variability as are all the natural environments related generic scrub factors. Alternatively, problems with the launch vehicle are specific to that launch vehicle family and would not be expected to vary on a monthly basis. If a launch vehicle is carrying crew on board, then that launch is subject to a whole slew of crew related launch scrub factors highlighted in blue in Table 1.

Table 1. Launch Scrub Factors

Category	Factor	Monthly Variability?
Generic	Launch Weather	Yes
	Range Weather	
	Winds Aloft	
	Tropical Cyclones	
	COLA's	
	Range Systems	No
	Range Conflicts	
	Range Intrusions	
Launch Vehicle Specific	Launch Vehicle	No
	Ground Systems	
	Human Error	
Mission Specific	Spacecraft	Yes
	Mission Systems	
	Crew Health	
	Crew Ingress	
	Crew Systems	
	Crew Actions	
	Abort Weather	
	Abort Sea State	

The probability of occurrence of each factor shown in Table 1 was estimated along with an associated delay duration probability distribution. The delay duration is the number of days between the scrub and when the next subsequent attempt can be made. A delay probability distribution is required if the historical data indicates that the duration of the delay is variable, which was found to be true in some cases.

The methodology used to develop the occurrence probabilities and associated delay distributions depended upon the nature of the scrub factor, the availability of public historical records or related previously published work, and whether or not simplifying assumptions could be made.

The launch attempt histories of the Space Shuttle, Atlas V, Delta IV and Falcon 9 launch vehicles were reviewed to determine scrub causes and scrub delay durations measured in days. Also if a mission launched late relative to its planned liftoff or T-0 time on a given launch date, then the reason for the delay and the duration of the delay in minutes was identified where possible. Only public sources were used, with Spaceflight Now's Launch Log being a particularly helpful source [1]. Likewise, papers authored by Mark Cleary and published by the Air Force Space Museum provided excellent information on the Atlas V and Delta IV launch attempts from Cape Canaveral Air Force Station [2]. Efforts were made to cross verify the data set using other public sources such as government and industry press releases. The launch attempt histories for Atlas V, Delta IV and Falcon 9 are shown in tabular form in the Appendix. The Space Shuttle launch attempt history can be found in a paper published in 2014 [3].

No proprietary data is presented in this paper or in any of the results shown. However, the model can be modified to utilize proprietary data with permission in order to run customer specific analyses.

3. GENERIC SCRUB FACTORS

Weather

Launch vehicle are typically restricted from being launched during adverse weather. These constraints are much more restrictive than those associated with commercial air flight. Launches are prohibited if there is a potential for natural lightning or launch induced lightning as past launch failures have been caused by lightning striking the launch vehicle [4]. From a safety perspective, there must be sufficient visibility to track the vehicle and determine if it has gone off course. If a launch vehicle goes off course it could threaten people outside the cleared launch danger area. The flight termination system needs to be activated in such a scenario. Low temperatures may need to be avoided based upon the design limits of the launch vehicle. Surface winds and low level winds must also be within the capabilities of the launch vehicle and launch complex for the critical moment of liftoff to prevent structural loads exceedances or the potential for the launch vehicle to strike the launch umbilical tower.

The work by Altino et al [5] on natural environments related probabilities for NASA's Space Launch System (SLS) launch vehicle was used as the basis of estimate for weather scrubs. That paper estimated the probability that the weather would not exceed natural environments constraints on launches such as the presence of lighting, thunderstorm activity, low ceilings, high winds, and low temperatures. The

probability of a launch being scrubbed due to these natural environments factors is shown in Figure 1. The figure indicates, for example, that on any given day in January that there is an approximate 30% chance that the weather will cause a scrub. The situation improves significantly outside of winter months.

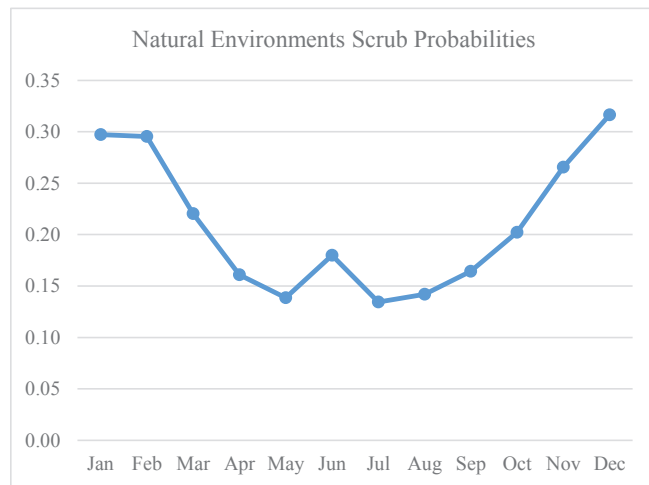


Figure 1. Space Launch System (SLS) Natural Environment Estimated Scrub Probabilities

Note that the Altino paper included hourly probabilities in addition to the daily averages. There can be considerable variability throughout the day. For example morning hours in the summer are significantly better than late afternoon hours.

However, the analysis being presented in this paper assumes that the launch hour is unknown. Therefore, the daily average scrub probability for the month as shown in Figure 1 were used. For a planetary mission analysis, the launch window time of day will be known so that the more accurate estimate for that particular hour can be used. Likewise for any launch, once the launch window is known the more accurate hourly probabilities for that month can be used.

Another key assumption is that estimates for SLS natural environments scrub probabilities are similar to other launch vehicles. This should be a reasonable assumption since the launch weather criteria for SLS are similar to other launch vehicles. The model can be updated to use launch vehicle specific information as it becomes available.

Scrubs for weather are assumed to have a one-day duration such that the simulation may allow an attempt the next day subject to other factors such as the ability of the launch team to support the attempt.

The natural environments analysis did not include winds aloft or tropical cyclone related launch scrubs.

Winds Aloft

Launch vehicles pass through the atmosphere at an increasing velocity as they ascend into orbit. During the flight through

the atmosphere they are subject to external forces from the wind profile that they pass through. While surface winds may be light and variable, the upper level winds may well be at or above hurricane force. Also the direction of the wind may be radically changing at different altitudes. The launch vehicle dynamic ascent guidance program must be capable of responding to these changing forces in order to avoid exceeding the structural limits of the launch vehicle. Weather balloons are released prior to a launch attempt so as to measure the winds at altitude and develop an upper level winds profile. If the winds are unfavorable, meaning that the structural limits are likely to be exceeded, then the launch will be delayed until the wind profile comes into accepted limits. If the window expires before this happens a scrub is called.

The estimate for the probability of scrubs due to unfavorable winds aloft was based upon a paper by Harrington et al regarding Space Shuttle day of launch trajectory design [6]. That work indicated that the likelihood of having favorable winds also varied on a monthly basis. Additionally the probability of scrub could be reduced by having an alternate trajectory design and even further reductions could be gained by having the ability to develop a trajectory design specific to the upper wind profile on the day of launch. The Space Shuttle Program called this capability day of launch initialization load updates or DOLILU. The scrub probability estimates are shown in Figure 2.

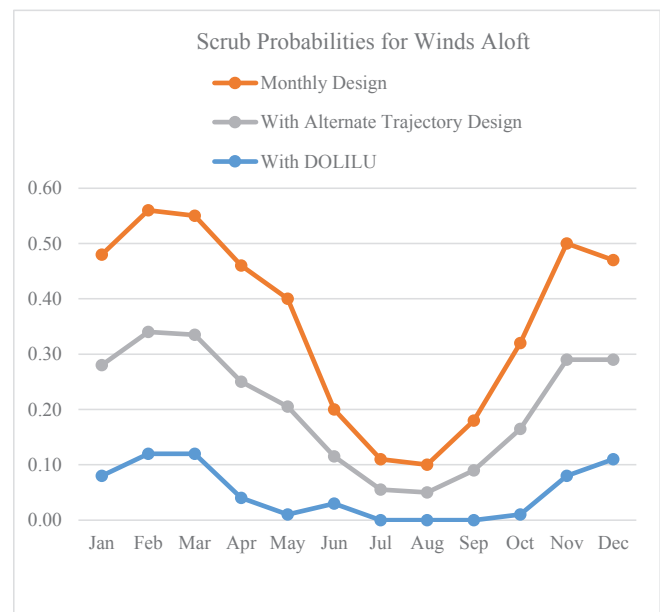


Figure 2. Shuttle Scrub Probabilities for Winds Aloft

Summer months are clearly better than other months and having a DOLILU capability is very beneficial. Note that these results are specific to the Space Shuttle. Any of the three Space Shuttle estimates can be used when analyzing other launch vehicles or alternatively launch vehicle provided estimates can be used. Scrubs for winds aloft are assumed to have a one-day duration subject to other factors that can constrain a launch attempt.

Tropical Cyclones

The methodology for determining the tropical cyclone scrub probability and delay duration basis of estimate is illustrative of the way many of the scrub categories were analyzed using empirical data. Tropical cyclones including tropical storms

and hurricanes can cause launch scrubs during hurricane season i.e., June through November. Although rare, tropical cyclones may occur in other months. The probability of a tropical cyclone causing a launch scrub was estimated using historical data from the Space Shuttle, Atlas V, Delta IV, and Falcon 9 launch attempt history as shown in Table 2.

Table 2. Tropical Cyclone Launch Scrub Basis of Estimate

	Number of Launch Attempts Per Month												Totals	
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
Space Shuttle	20	19	16	31	15	20	23	21	17	25	28	20	255	1981
Atlas V (LC-41)	6	5	7	5	8	5	2	10	3	1	4	3	59	Through
Delta IV (LC-37)	5	1	8	0	10	4	8	6	0	1	7	9	59	March
Falcon (LC-40)	4	4	2	2	3	5	1	2	3	1	2	6	35	2015
	35	29	33	38	36	34	34	39	23	28	41	38		
Number of Hurricane Related Launch Scrubs Per Month														
Space Shuttle								1	1	1			3	
Atlas													0	
Delta						1							1	
Falcon													0	
	0	0	0	0	0	1	0	1	1	1	0	0		
Empirical Hurricane Scrub Probabilities	0.000	0.000	0.000	0.000	0.000	0.029	0.000	0.026	0.043	0.036	0.000	0.000		
Hurricane Related Launch Scrubs (with June scrub divided between June and July)														
	0	0	0	0	0	0.5	0.5	1	1	1	0	0		
Modified Hurricane Scrub Probabilities	0.000	0.000	0.000	0.000	0.000	0.015	0.015	0.026	0.043	0.036	0.000	0.000		
Delay durations (days)						1		1	8	5				
Delay distribution	DISC(0.5,1, 0.75,5, 1.0,8, Hurricane Delay Seed)													

Launch attempts were categorized by the month in which they occurred and then tropical cyclone related delays that occurred during the launch countdown period were identified. There were 4 identified launch scrubs due to tropical cyclones—one each in June, August, September and October. The June data point was divided into two half points, one in June and the other in July because the resulting probabilities became more consistent with tropical cyclone climatology. The climatological record for the Atlantic Ocean indicates tropical cyclone activity has a relatively low probability of occurrence during the June and July timeframe ramping up to a peak probability of occurrence in the September-October timeframe [7].

If a tropical cyclone scrub occurs in the simulation model, the next launch attempt is delayed for some number of days. The number of days is based upon a distribution created from the historical data, i.e., two 1-day delays, one 5-day delay and one 8-day delay. The nomenclature representing this discrete

empirical distribution is shown in Equation 1. DISC means discrete empirical distribution with paired numbers indicating first the cumulative probability and second the delay duration. The term “Seed” in Equation 1 is used to specify the random number seed so that a specific string of random numbers from the simulation’s random number generator can be assigned to determining the delay duration.

$$\text{DISC} (0.5,1, 0.75,5, 1.0,8, \text{Seed}) \quad (1)$$

This construct allows the discrete event simulation to determine the duration consistent with the empirical record using independent and identically distributed random numbers ranging between 0 and 1. In this case 50% of the tropical cyclone delays will manifest themselves as 1day, 25% will be 5 days and 25% will be 8 days.

Note that there are several other instances of tropical cyclones causing launch delays but most occur prior to the start of the launch countdown period. The historical data analysis

assumed that the launch countdown period began on launch minus three days. This was because the Space Shuttle countdown procedure began on that day. No launch attempts have been scrubbed on the day of an attempt due to a tropical cyclone. Consequently, when starting the simulation at the start of propellant loading the tropical cyclone scrub probability would be zero.

Collision Avoidance (COLA)

Assessments are provided by the Air Force to ensure that a vehicle being launched will not put its spacecraft on a collision course with on-orbit satellites or trackable orbital debris. These assessments are performed on the day of the launch and may result in collision avoidance launch window cutouts known as COLAs. Typically they will last for a few minutes. The presence of a COLA in a launch window will rarely cause a scrub. It will be dependent upon the duration of the planned window with most launch windows typically being longer than the COLA. But such will not always be the

case, especially with missions having instantaneous or very short duration launch windows of only a few minutes.

The historical data, the resulting COLA delay probabilities and associated delay distributions are shown in Table 3. Only one COLA that resulted in a scrub was identified. This was related to the ASAT test performed unexpectedly by China resulting in a 13-day delay to the launch of an Atlas V. This was categorized as a rare event but nonetheless include in the model. Six cases were identified in which the presence of a COLA resulted in a delay to the planned T-O time or liftoff of a launch. The occurrence probability of a COLA that causes a T-O delay was estimated using that data as shown in Table 3. The simulation model is programmed to check for the occurrence of a COLA delay based upon that probability and then if there is a delay determine the duration of the delay using the historical delay distribution shown at the bottom of Table 3. If the delay is greater than the launch window, the model will implement a scrub with a duration of 1 day.

Table 3. COLA Scrub & Delay Probabilities & Delay Distributions

	STS	Atlas V	Delta IV	Antares	Falcon	Sum	Sum
T-0 Delays	4	0	2	0	0	6	
Scrubs	0	1	0	0	0		1
Attempts	187	74	64	14	36	375	375
Probability	0.0214	0.0000	0.0313	0.0000	0.0000	0.0160	0.0027
Delay	13 minutes ~10 minutes	13 days China	1 minute 1 minute			COLA Delay Occurrence Probability	Rare Event Occurrence Probability
Durations	6 minutes 2 minutes	ASAT Test					
Delay Distribution: (days)	DISC(0.33,1, 0.50,2, 0.67,6, 0.83,10, 1.0,13, COLA Duration RN Seed)						13

Range Conflicts

An entity known as the Eastern Range (here after “the Range”) has been established to oversee launches that occur from the East Coast of Florida and ensure the safety of the public. The Range supports launch service providers known as users including United Launch Alliance (ULA) with their Delta and Atlas launch vehicles and Space Explorations Technologies Corp. (SpaceX) with their Falcon launch vehicle. In the future NASA’s Space Launch System (SLS) will also use the Range.

In order to accommodate the various users in an equitable manner the Range generally operates on a first come, first served basis. Each user may make a reservation on the Range to launch a mission. A mission may only have one date reserved on the Range. The Range typically requires at least 2 days between reserved launch dates since only one launch can be supported on any particular date and at least a day between launches is needed to reconfigure Range systems for the next user. In other words the Range could be reserved today for an Atlas V launch and two days later for a Falcon 9 launch.

The Range limitation of being able to support only one user at a time can lead to conflicts when launch scrubs occur. Suppose an Atlas V has the Range today and the Falcon 9 has the Range reserved for two days from now. If the Atlas V launch attempt is scrubbed, the Range will allow ULA to make an attempt the next day. When that happens, the Falcon 9 launch reservation will be slipped one day. Table 4 shows the probability of occurrence and associated delay duration for Range conflicts. These particular Range related delays to launch would only be part of the launch probability estimate if the simulation is begun at the launch minus 2 days point or earlier.

There are other ways in which the Range capabilities and presence of other Range users works its way into the cumulative probability of launch metric. Suppose one is particularly interested in the cumulative launch probability of a hypothetical Atlas V with an initial launch attempt of today that has just scrubbed. The Range allows it to try to launch again tomorrow, but if the launch scrubs again, the ULA team would typically have to give up the Range and allow the SpaceX launch team to take their turn with their Falcon 9.

Table 4. Range Conflict Delays to Launch

	STS	Atlas V	Delta IV	Falcon 9	Sum
Initial Launch Attempts	135	53	29	16	233
Range Delays To Initial Attempt	3	1	0	0	4
Delay Durations (Days)	1	1			0.0172
Delay Distribution:	DISC(0.75,1, 1.0,4, Range_Conflict_RN_Seed)				Delay Probability

Our hypothetical Atlas V mission will not be able to get back on the Range until after the Falcon 9 takes its turn on the Range. And theoretically there could be other users on the Range with reserved dates that could bump the Atlas V out even further. During the Space Shuttle era one Shuttle mission had to wait nearly three weeks before it could get back on the Range.

Note that the general Range sharing rules might not apply if the particular mission trying to launch is of significant national importance. That mission would be allowed to make multiple attempts regardless of other users being in the queue. The simulation has a feature that allows the Range sharing rules to be ignored.

Supposing there was no Falcon 9 or any other user with a Range reservation after the Atlas V mission scrubbed twice in a row, then the ULA launch team could continue making attempts. However, the number of attempts in a row is limited by crew rest requirements for both the ULA and Range launch team, as well as the capabilities of the supporting ground infrastructure and propellants. The Range will typically want to stand down for a day or two after supporting three attempts in a row over a three day period.

All of these Range reservation rules are subject to a certain amount of negotiation depending upon the needs and desires of all the parties, i.e., the Range and the users in conflict. The simulation is coded to account for the Range reservation interactions but there is also a switch in the model to ignore those rules. This provides the capability to see how much the cumulative probability of launch can be improved upon by making an agreement with the Range to block out multiple days in a row.

Range Systems

The Range provides authority to launch from a public safety perspective. Many resources including communications systems, radars, optical tracking devices, flight termination

systems, weather monitoring systems along with command, control and range safety officers are required to perform this critical service. There are launch commit criteria that must be met so if a critical Range Systems resource goes down then the Range will go into a Red status until the problem can be resolved. Some of these problems can result in a launch scrub. The historical data summary for Range Systems related scrubs of 1 or more days is shown in Table 5.

Table 5. Range Systems Scrubs

	STS	Atlas V	Delta IV	Antares	Falcon	Sum
Scrubs	2	3	0	0	1	6
Attempts	187	74	64	14	36	375
Probability	0.0107	0.0405	0.0000	0.0000	0.0278	0.0160
Delay Durations (days)	1	1			1	Scrub Probability
Delay Distribution: (days)	DISC(0.86,1, 1.0,42, Range Systems RN Seed)					

Range Intrusions

In order to ensure public safety the Range will make every effort to keep aircraft, boats, cars and individuals out of the launch danger areas. Notices to airmen, mariners and public announcements are made for the day and the launch period for specific areas to be avoided. If there is a Range intrusion, the Range will go to a Red status and the launch attempt will have to hold until the intruder has cleared the danger area.

The historical data summary for Range Intrusions related T-0 delays in minutes is shown in Table 6. Note that these are delays to the planned T-0 time and in most cases will not result in a launch scrub. Similar to how COLA delays are modeled, the simulation will check for the occurrence of an intrusion, then determine the duration of the intrusion, and a scrub will be implemented only if the intrusion duration exceeds the launch available launch window.

Table 6. Range Intrusion Summary Data

	STS	Atlas V	Delta IV	Antares	Falcon	Sum
Delays	11	3	1	1	1	17
Attempts	162	74	64	14	36	350
Probability	0.0679	0.0405	0.0156	0.0714	0.0278	0.0486
Delay Durations (Minutes)	7	13	12	30	30	Intrusion Probability
	55	30				
	30	23				
	3					
	1					
	10					
	1					
	10					
	10					
	30					
Delay Distribution: (minutes)	DISC(0.18,1, 0.24,3, 0.29,7, 0.46,10, 0.53,12, 0.59,13, 0.65,23, 0.94,30, 1.0,55, Intrusion Duration RN Seed)					

4. LAUNCH VEHICLE SCRUB FACTORS

Launch vehicle scrub factors are those associated with the launch vehicle flight hardware, the supporting infrastructure and human errors that may occur during the conduct of the countdown by the launch team. For purposes of this paper launch vehicle flight hardware and ground systems scrubs are combined into one category for each of the three launch vehicle families, i.e., Atlas V, Delta IV and Falcon 9. All vehicle variants were lumped together into each of these families. The data sets found in the Appendix show all launch scrubs along with their associated delay duration and reason for the scrub. Those scrubs that are not specific to the launch vehicle or directly associated ground systems were removed to create the estimated probability of scrubbing for a launch vehicle related problem.

Figure 3 provides the historical proportion of scrubs along with a 95% confidence interval for the proportion determined using the exact binomial confidence interval technique described by Morisette and Khorram [8]. Note the varying sizes of the confidence intervals. Theoretically, the true scrub probabilities reside within the confidence intervals. However, due to the evolutionary nature of launch vehicles, particularly the Falcon 9 family, future launch scrub probabilities are somewhat problematic to predict even with this degree of uncertainty. As more launches occur the historical proportions will move up or down depending upon the launch countdown performance and the confidence intervals will become smaller.

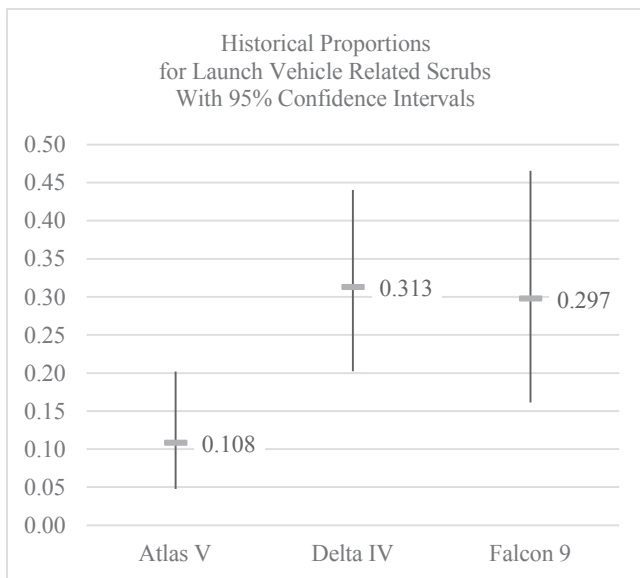


Figure 3. Launch Vehicle Scrub Probability Estimates

The scrub duration distributions in terms of numbers of days are shown in Table 7. For both Atlas V and Delta IV the most likely scrub duration has been 1 day, whereas for Falcon 9 the most likely duration has been 3 days.

Table 7. Launch Vehicle Scrub Duration Distributions (Days)

Atlas V	DISC(0.625,1, 0.75,2, 0.875,9, 1.0,17 RN_Seed)
Delta IV	DISC(0.60,1, 0.75,2, 0.80,3, 0.85,4, 0.90,9, 0.95,24, 1.0,40, RN_Seed)
Falcon 9	DISC(0.27,1, 0.64,3, 0.82,4, 0.91,12, 1.0,22, RN_Seed)

The launch vehicle related probability of a delay to the planned T-0 time is shown in Figure 4. The confidence intervals are even wider due to the smaller data set since this metric is based upon actual launches as opposed to all launch attempts regardless of outcome. There is considerable overlap of the confidence intervals.

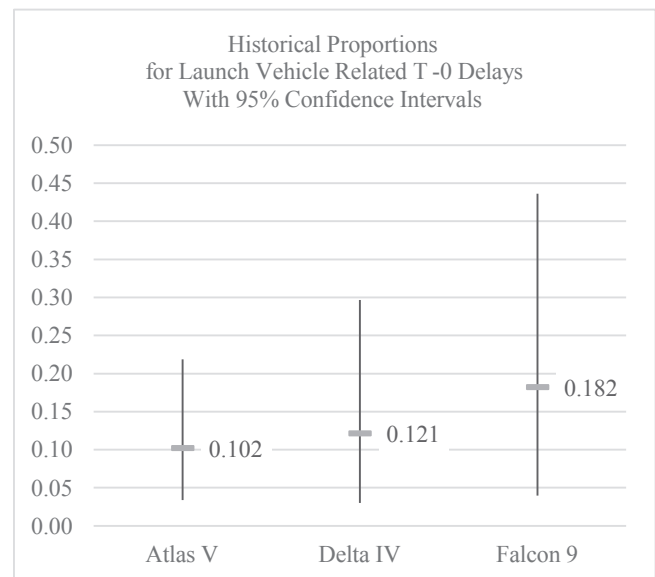


Figure 4. Launch Vehicle T-0 Delay Probability Estimates

Delays to the T-0 time can translate into a launch scrub in the simulation if the duration of the delay exceeds the available launch window. Distributions for the durations of the T-0 delays in minutes based upon the flight history of each of the three launch vehicles are shown in Table 8.

Table 8. Launch Vehicle T-0 Delay Duration Distributions (Minutes)

Atlas V	DISC(0.1,1, 0.2,2, 0.3,4, 0.5,13, 0.6,25, 0.7,33, 0.8,51, 0.9,97, 1.0,147, RN_Seed)
Delta IV	DISC(0.25,11, 0.50,55, 0.75,106, 1.0,134, RN-Seed)
Falcon 9	DISC(0.25,100, 0.50,113, 0.75,156, 1.0,225, RN_Seed)

Human Errors

The Space Shuttle Mission Summary [9] identified four cases in which launch scrubs occurred due to human errors on the part of the launch team. These included: sending the wrong

commands to the vehicle during propellant loading (STS-61C); failure to service the hydraulic system (STS-73); scrub for a pip pin discovered on the launch pad (STS-92); and incorrectly interpreting readings from a hydrogen leak detection system (STS-93). The probability of a scrub due to human error for the Space Shuttle was calculated by dividing the 4 scrubs by the 255 Space Shuttle launch attempts, yielding a scrub probability of 0.0157. Three of the scrub durations were of the 1-day variety with one instance requiring 2 days to correct.

The Atlas V, Delta IV and Falcon 9 historical data sets do not appear to have similar examples but human errors could be at the root of some of the scrubs. The simulation includes an option to include Human Error scrub risk based upon the Space Shuttle historical data if one wants to take a conservative approach.

5. MISSION SPECIFIC FACTORS

Mission specific scrub factors include the spacecraft or payload being launched and the attendant mission control systems.

Spacecraft

Six instances in which the spacecraft caused a launch to be scrubbed were identified. These were lumped together since the payload is typically an independent entity relative to its launch vehicle. The Antares launch attempt history was also reviewed and included. The resulting spacecraft scrub probability and associated delay distribution are shown in Table 9. An additional case was identified in which the spacecraft, while not being the direct cause of scrub, did extend the duration of the scrub by a large amount. This was the GOES-N spacecraft aboard a Delta IV. The August 16, 2005 attempt was scrubbed due to a launch vehicle battery problem, but the 282 day duration of the scrub was driven by GOES-N. To account for this type of launch scrub duration risk, a 282-day delay was added to the delay durations in Table 9.

Table 9. Spacecraft Related Scrubs

	STS	Atlas V	Delta IV	Antares	Falcon	Sum
Payload Scrubs	2	0	3	0	1	6
Attempts	255	74	64	14	38	445
Probability	0.0078	0.0000	0.0469	0.0000	0.0263	0.0135
Delay Durations (Days)	5		2		8	
	5		1			
			1			
Delay Distribution	DISC(0.29,1, 0.43,2, 0.71,5, 0.86,8, 1.0,282, Spacecraft Delay RN Seed)					

Mission Systems

For the purposes of this paper mission systems refer to the spacecraft's ground communication and control systems after it is launched. For example, mission control for the Space

Shuttle resided at the Johnson Space Center (JSC) in Texas whereas launch control was conducted at the Kennedy Space Center in Florida. Because JSC took over flight control upon liftoff at T-0, all required mission systems needed to be operational prior to launch.

No launch vehicle scrubs due to Mission Systems separate from the launch control systems were identified. However, three Space Shuttle launches experienced a T-0 delay due to problems associated with mission systems. The probability of a mission systems delay using just the Space Shuttle data set was estimated at 0.0132 with a 95% exact binomial confidence interval of from 0.0016 to 0.0458. The historical durations of the T-0 delays were 14, 23, and 97 minutes and those data were used to develop a discrete empirical distribution for the simulation.

The simulation can be run with or without the mission systems risk being active. When it is active the simulation will first see if there is a mission systems T-0 delay and if so determine the duration of the delay. If the delay exceeds the available launch window then a scrub occurs. The duration of the scrub would be at least 1 day subject to other factors that determine when the next launch attempt could be made.

6. CREW SPECIFIC FACTORS

Crew specific factors relate to the health of the crew for the flight, getting the crew on board and their actions during countdown and the additional weather constraints associated with contingency aborts. The Space Shuttle historical data were primarily relied upon to develop scrub probabilities and scrub delay durations.

Crew Health

Astronauts flying on the Space Shuttle entered a health stabilization program [10] prior to flight to reduce the chance that an illness would delay a mission. The program seems to have been quite effective as only launch was scrubbed due to a crew health concern. This occurred at the Mission Management Team (MMT) tanking meeting where NASA decides whether or not to load the Space Shuttle with propellants. The likelihood of occurrence is based upon the one event divided by the 233 times the MMT tanking meeting took place yielding a probability of 0.0043. On average 6 astronauts flew on each Space Shuttle mission. This works out to a 0.0007 probability per crew member of being ill at the time of the MMT tanking meeting. The Space Shuttle experience was a 2-day delay until the next launch attempt was made.

Crew Ingress

During the Space Shuttle era the astronauts entered the orbiter after propellant loading of the External Tank was completed. The process of getting all the astronauts inside and then closing out the crew cabin and launch pad for launch was a highly choreographed activity.

The Space Shuttle historical record [3] shows that the launch countdown got past tanking 187 times such that crew ingress could begin. There were times in which problems occurred delaying the completion of crew ingress. Five of these problems resulted in delays to the planned liftoff time and two problems resulted in launch scrubs. Table 10 shows the resulting probabilities for scrubs, T-0 delays and the T-0 delay duration distribution. Similar to how other T-0 delay risks are handled, the simulation will first check if a T-0 delay has occurred, determine the duration of the delay using the empirical discrete distribution and then cause a scrub if the T-0 delay is greater than the available launch window. The duration of the scrub will be 1 day subject to potential further delay depending upon other scrub factors and the capability of the launch team to support a next-day attempt.

Table 10. Crew Insertion Scrub and Delay Data

Crew Insertion Initiated	Scrubs	T-0 Delays
187	2	5
Scrub Probability	0.0107	
Delay Probability		0.0267
	Delays (Days)	Delays (minutes)
	1	3
	1	5
		10
		14
		21
		50
Distribution for T-0 Delays		
DISC(0.167,3, 0.333,5, 0.500,10, 0.667,14, 0.833,21, 1.0,50)		

Crew Systems

The crew launch and entry partial pressure suits worn by the astronauts are the primary crew systems being addressed by this risk factor. During the Space Shuttle era there was one launch in which the planned T-0 time was delayed by 98 minutes to replace fuses in two suits. Partial pressure suits were worn by all crew members after the Challenger accident, which happened to be the launch that was delayed to replace the fuses. The Space Shuttle historical record [3] indicates that 150 launch attempts got to the T-9 minute point. Using an average crew size of 6 yields a risk exposure of 900 suits of which 2 had a problem resulting in a launch T-0 delay. This indicates a 0.0022 probability of a crew system failure per crew member. A crew systems failure is assumed to result in a 98 minute delay to the planned T-0 time based upon the one historical data point. This means that a scrub will occur in the simulation only if the available launch window is less than 98 minutes. The duration of the scrub will be 1 day subject to other factors.

¹ SWH is equal to the mean height of the highest one third of the waves in the area being observed.

Crew Actions

During the Space Shuttle era there were critical actions that the crew performed during the last 9 minutes of the countdown in order to support launch. The crew was directly involved in the starting of the Auxiliary Power Units (APUs) at approximately T-5 minutes and counting. The NASA Space Shuttle Missions Summary [9] identifies 2 instances of a crew action causing a delay. One was a 48 second delay, which was rounded to 1 minute, and the other was a 5 minute delay. The 5 minute delay resulted in a launch scrub. There were 152 times in which the countdown got to the T-5 minutes and counting point that required crew action for proceeding. The probability of a crew action delay is thus 2 divided by 152 equaling 0.0132. The delay duration in minutes would be either 1 minute or 5 minutes with equal likelihood based upon the historical record. A scrub would only occur if the delay duration is greater than the launch window.

Abort Weather and Abort Sea State

The Space Shuttle concept of operations included two abort scenarios that had weather related launch commit criteria. The first was the return to landing site (RSLs) abort mode, which meant making sure that the weather at the runway on the Kennedy Space Center was within limits for a returning Space Shuttle orbiter. The runway is approximately 4 miles from the launch pad and the weather criteria for landing were different than for launch. Although the weather at each location is highly correlated, the Space Shuttle launch historical paper [3] and Space Shuttle Mission Summary [9] show that there were times in which launch was scrubbed solely due to weather conditions at the runway. The other abort scenario was the Trans-Atlantic-Landing (TAL). The TAL landing sites were in Europe and Africa. Prior to launch at least one of the TAL sites needed to have acceptable weather conditions. Here again there were times when Space Shuttle launches were scrubbed or delayed due to TAL weather conditions.

The situation will be quite different for future U.S. human space flight missions. The spacecraft currently being developed—Orion, Dragon, and CST-100—are all capsules with abort concept of operations that would have them splashdown in the Atlantic Ocean in the event of a launch vehicle ascent mishap. Consequently, it will be the sea state conditions in the ocean abort zones that will be of interest. While the sea state conditions in the abort zone include winds, temperatures, and clouds, the rest of this discussion is focused on wave heights and a metric called significant wave height (SWH).¹

During the Constellation Program, NASA studied abort constraints such as significant wave height levels that might be imposed before committing to launch. The probability that the wave heights would be less than various levels—3, 4

or 5 meters—during each month of the year for missions going to the International Space Station (ISS) was estimated by Barbré and Keller [11]. Their results are graphed in Figure 5.

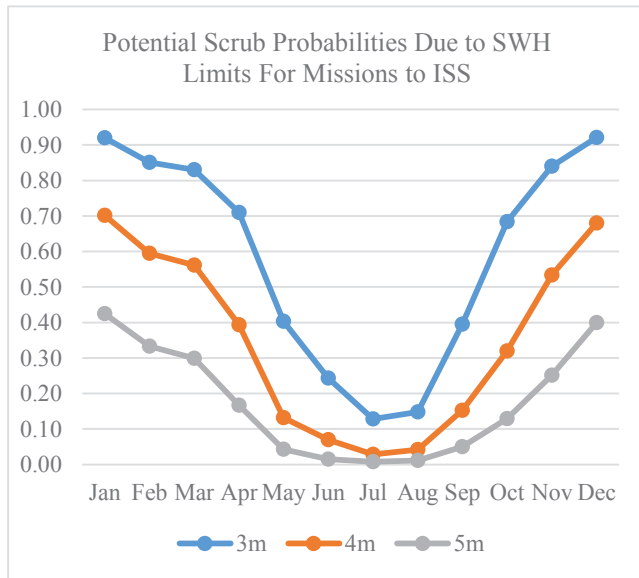


Figure 5. Abort Sea State Scrub Probabilities (ISS)

These missions have abort landing zones in the North Atlantic because they have to fly an ascent trajectory to match the ISS which is in a 57 degree inclination orbit. The published Constellation study information was used to estimate the potential probability of a scrub due to the waves exceeding threshold values. Figure 5 shows large monthly variability in scrub probability.

Assuming a 4-meter SWH limit, the probability of a scrub on any given day during the month of January would average approximately 0.70. The situation improves dramatically in the summer months. Raising the threshold values to 5 meters would provide improvement whereas lowering the threshold to 3 meters would increase the probability of a scrub. The simulation model treats a scrub due to abort sea states as a 1-day scrub and will try to launch again the next day subject to the other constraints and factors in the model.

7. DESCRIPTION OF ALPS

The Aerospace Launch Probability Simulation is a discrete event simulation model created using Arena by Rockwell Automation [12]. Textbooks on discrete event simulation and simulation modeling and analysis methods include those by Nicol et al [13] and Law [14].

The model is integrated with Excel for model inputs and outputs. Figure 6 shows a high level diagram of ALPS. The scenario parameters include the launch vehicle type, the payload type, the planned launch date, the duration of the

launch window, and whether or not crew are on board and if so how many. Rather than set a fixed planned launch date, ALPS can also be run with the planned launch date uniformly spread throughout the year so as to get an annualized average probability distribution. The Excel input worksheet also includes a listing of which risk factors are active and their associated scrub probabilities. The scrub delay duration distributions are resident within the Arena model. The simulation writes the results to an Excel file where the raw data are used to develop output metrics such as a cumulative probability distribution.

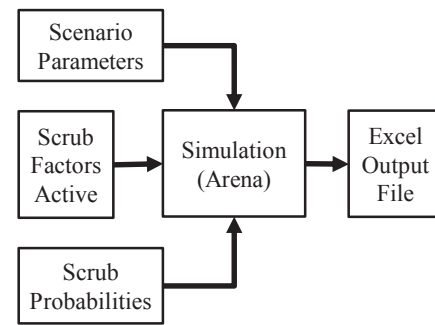


Figure 6. ALPS Overview

Figure 7 shows a screen shot from the simulation that helps illustrate how ALPS works. Across the top of the screen are placeholders for the planned launch date, the spacecraft, the launch vehicle and the launch window values. An entity representing the mission being launched is created and routed through the model checking for the occurrence of scrubs for each of the factors in the model.

Generally speaking the entity moves from left to right starting in Row 1 with the wet dress rehearsal if that risk factor is active. Otherwise the entity will generally start with checking for Range availability the day prior to the planned launch date. The launch minus one day check also includes the tropical cyclone and crew health risk factors.

If the Range is available and there are no scrubs due to tropical cyclones or crew health, then the entity will proceed to the day of launch. The remainder of Row 1 active risk factors are checked for scrubs and the Row 2 active factors are also checked. If multiple factors indicate a scrub, then the simulation will move forward in time the number of days associated with the greatest duration scrub. The next launch attempt may occur at that point so long as the Range is available. A separate sub model simulates the presence of other users on the Range.

The simulation can be run for any number of replications, with each replication representing one possible manifestation of when launch occurs relative to the planned launch date. The only difference between the replications is the random numbers used to drive the various risk models.

Atlas V reaches the 0.95 value at approximately the two week point. Further analysis is required to determine if there are statistically significant differences between the three launch vehicle CDFs.

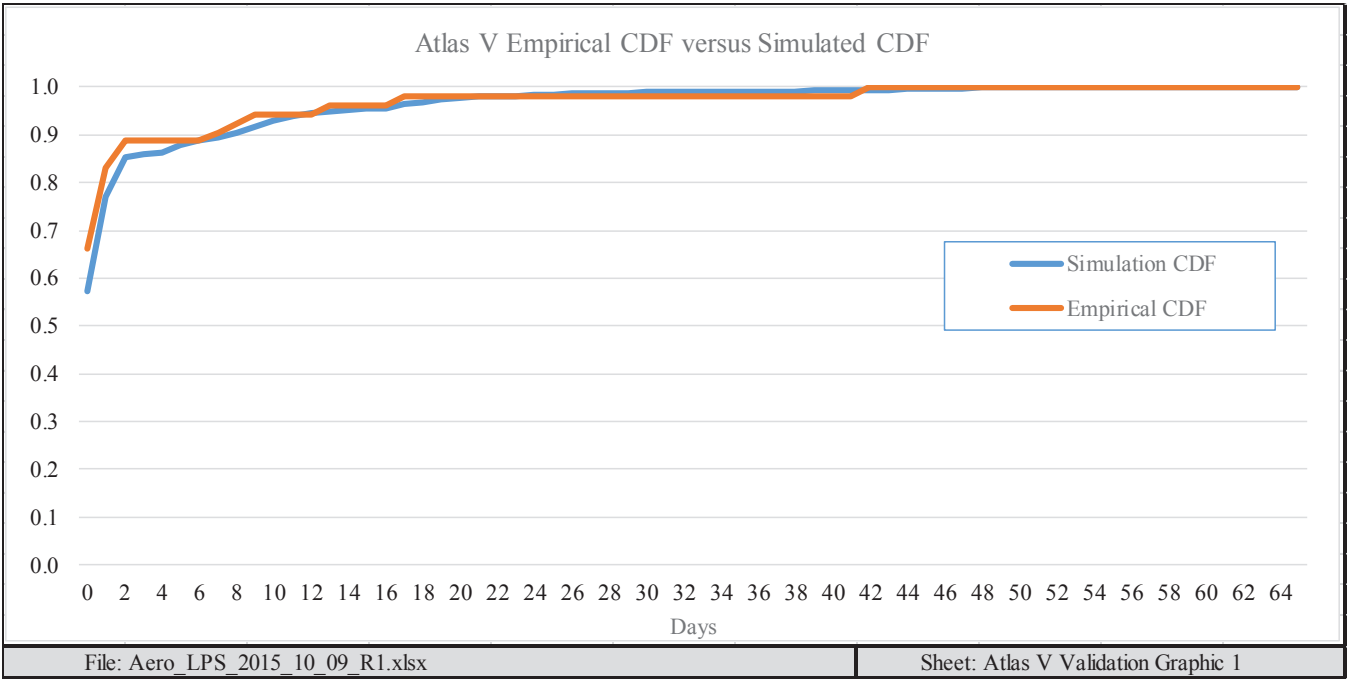


Figure 8. Atlas V Simulated and Empirical Cumulative Launch Probability Distributions

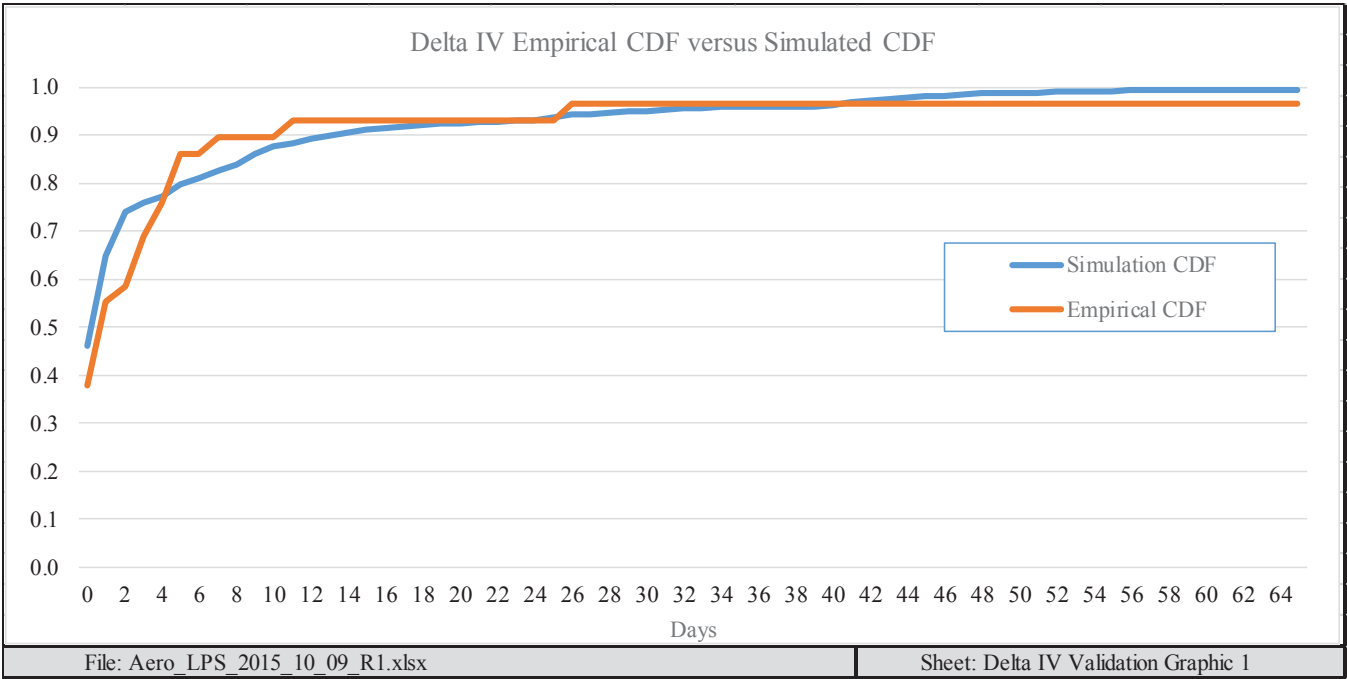


Figure 9. Delta IV Simulated and Empirical Cumulative Launch Probability Distributions

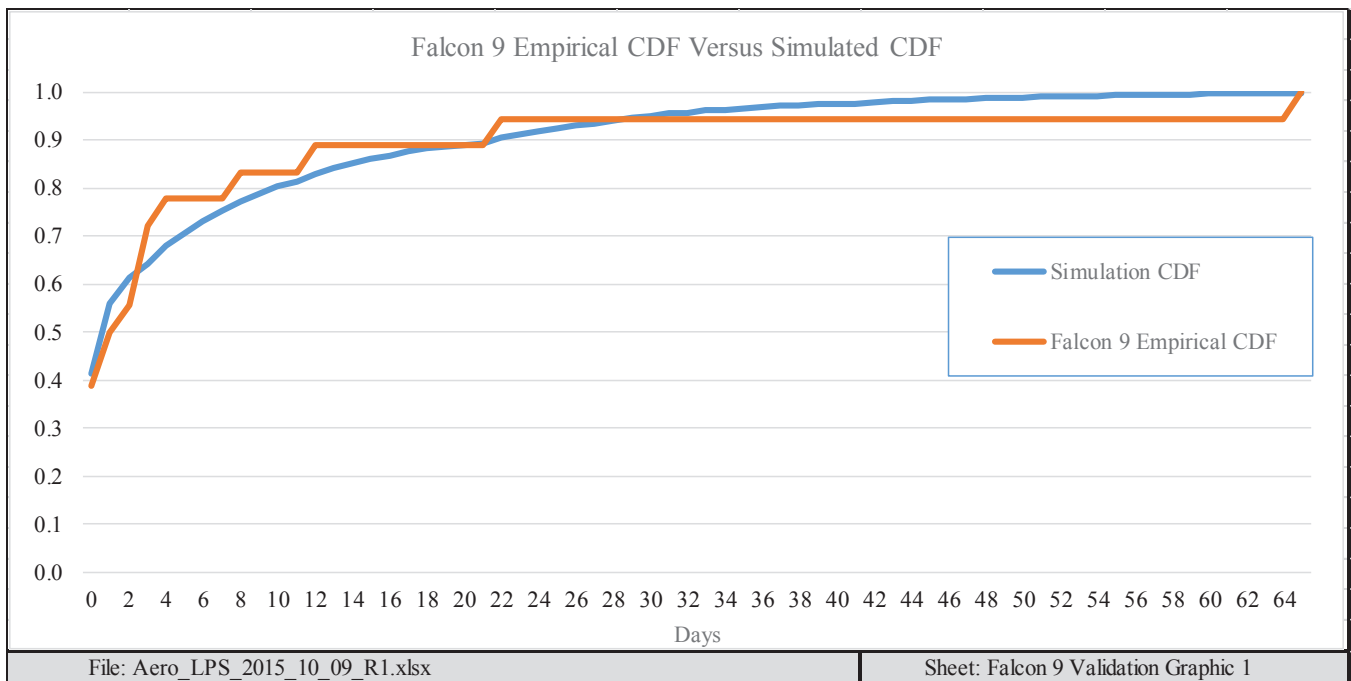


Figure 10. Falcon 9 Simulated and Empirical Cumulative Launch Probability Distributions

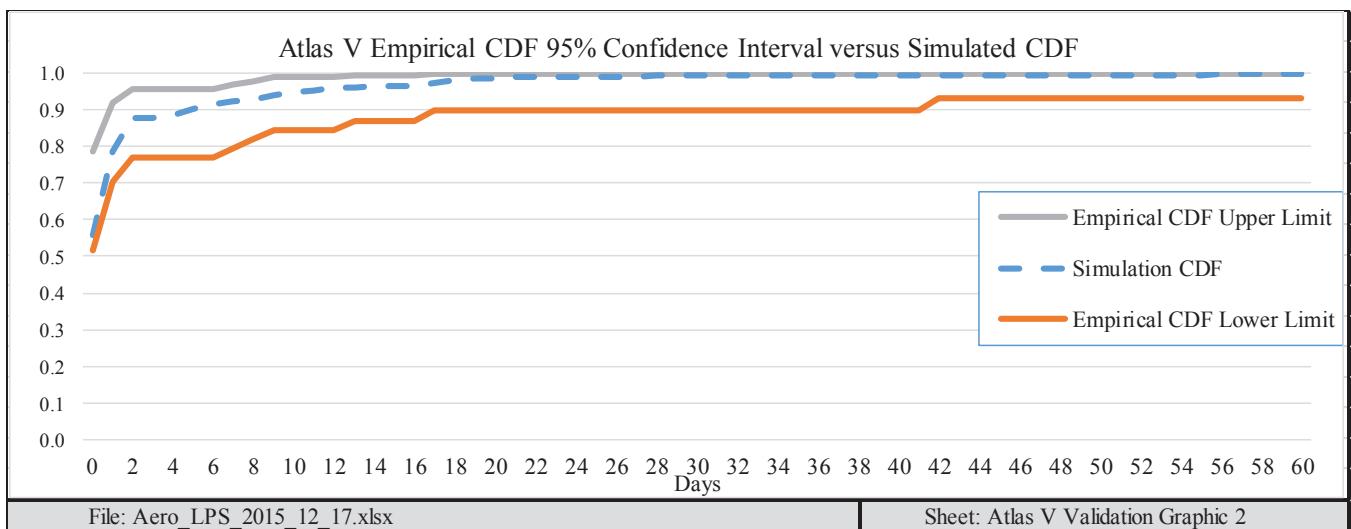


Figure 11. Atlas V Confidence Intervals (Empirical Versus Simulation CDFs)

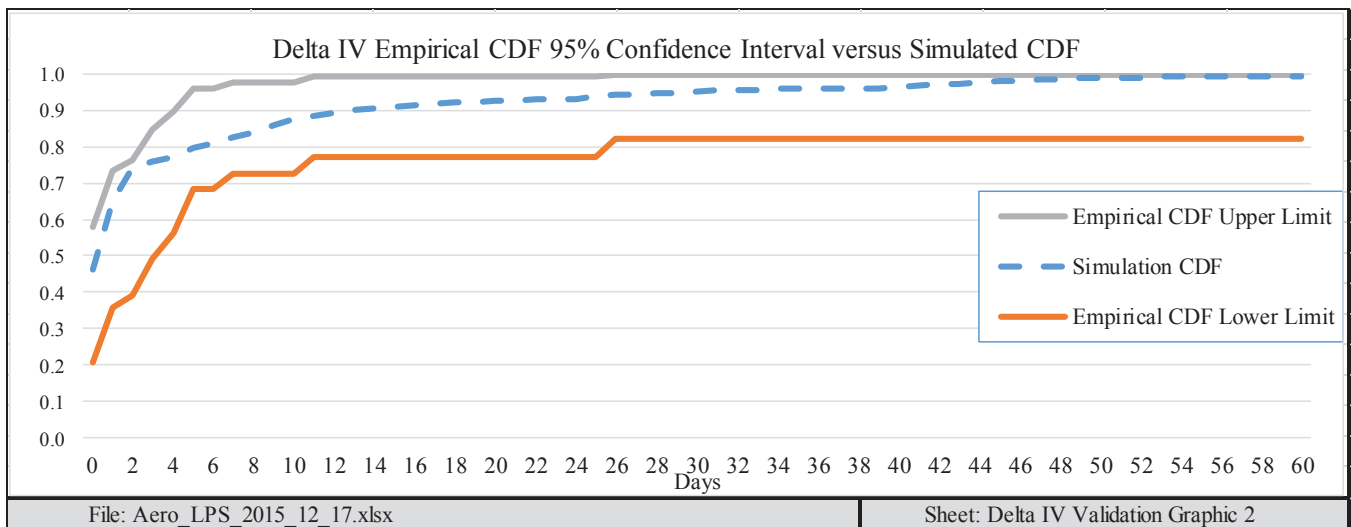


Figure 12. Delta IV Confidence Intervals (Empirical Versus Simulation CDFs)

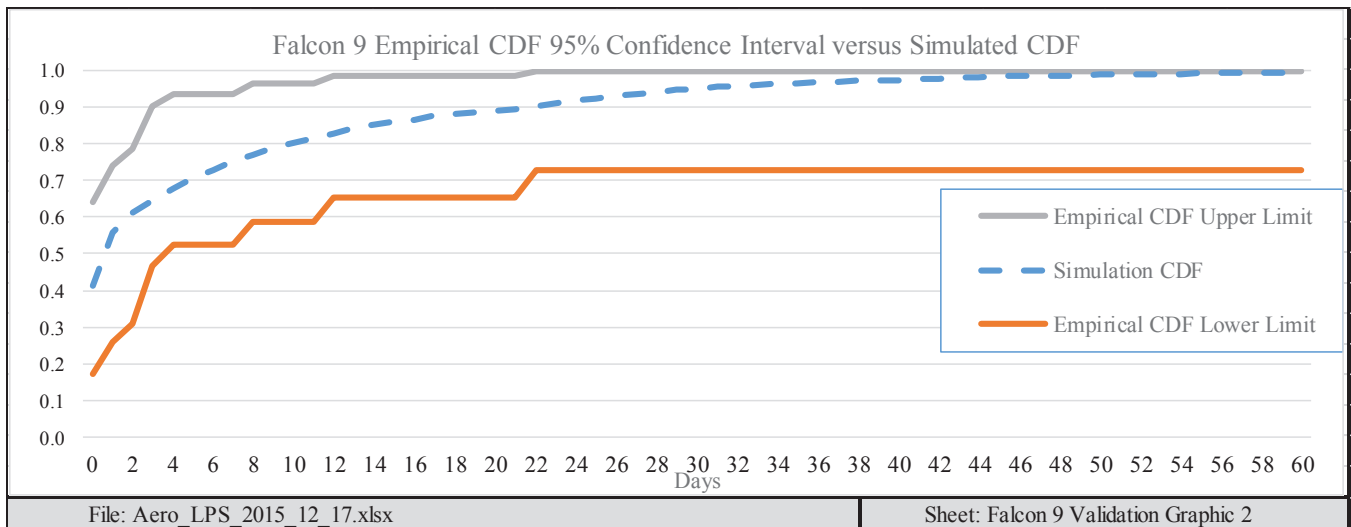


Figure 13. Falcon 9 Confidence Intervals (Empirical Versus Simulation CDFs)

9. CONCLUSIONS AND FORWARD WORK

The results of the Aerospace Launch Probability Simulation are shown to have validity for the set of circumstances analyzed. The ALPS generated cumulative launch probability distribution functions are consistent with the empirical data. ALPS can be used going forward to provide cumulative launch probability assessments for the Atlas V, Delta IV and Falcon 9 launch vehicles so long as it is kept up to date with the available historical data and configured appropriately for the specific mission and launch vehicle under study. ALPS can be used to help ensure mission success by exploring ways to improve launch probability. The simulation can be tailored to analyze mission specific scenarios such as planetary launch missions. It can also be used to estimate cumulative launch probability distributions for crewed flights.

Several forward activities are planned. Updating the historical records as launches occur will continue. The capability to discern between Cape Canaveral launches off the East Coast and West Coast launches from Vandenberg Air Force Base will be added. The model will be expanded to analyze other launch vehicles such as Antares and the Space Launch System. A related future effort is to incorporate the ALPS code into other simulations so that assessments of multiple launches can be enhanced. There may be an increasing COLA launch scrub or T-O delay risk associated with the increasing congestion in earth orbit. This is an area that requires future scrutiny. This paper does not address the complexity of each spacecraft and how that might influence the scrub probability. The spacecraft scrub probability for a specific mission under study might need to be revisited.

APPENDIX: LAUNCH VEHICLE SCRUB AND T-0 DELAY HISTORY

Atlas V Launch Attempt History								
#	Attempt Date	Launch Site	Outcome	Scrub Duration	Scrub Category	T-0 Delay?	T-0 Delays (minutes)	T-0 Delay Reason
1	21-Aug-02	LC-41	Launched					
2	12-May-03	LC-41	Scrubbed	1	Centaur	No		
3	13-May-03	LC-41	Launched			Yes	13	
4	17-Jul-03	LC-41	Launched			Yes	25	
5	17-Dec-04	LC-41	Launched			Yes	147	
6	10-Mar-05	LC-41	Scrubbed	1	FTS & Centaur	No		
7	11-Mar-05	LC-41	Launched					
8	11-Aug-05	LC-41	Scrubbed	1	Weather	No		
9	12-Aug-05	LC-41	Launched					
10	17-Jan-06	LC-41	Scrubbed	1	Weather	No		
11	18-Jan-06	LC-41	Scrubbed	1	Ground Systems	No		
12	19-Jan-06	LC-41	Launched			Yes	52	Weather
13	20-Apr-06	LC-41	Launched					
14	8-Mar-07	LC-41	Launched			Yes	33	Range systems; Range Intrusion; Launch Vehicle
15	14-Jun-07	LC-41	Launched					
16	10-Oct-07	LC41	Scrubbed	1	Centaur	No		
17	11-Oct-07	LC41	Launched					
18	10-Dec-07	LC-41	Launched			Yes	1	Unspecified
19	29-Feb-08	VAFB SLC-3E	Scrubbed	13	COLA (China Satellite)	No		
20	13-Mar-08	VAFB SLC-3E	Launched					
21	14-Apr-08	LC-41	Launched					
22	17-Mar-09	LC-41	Scrubbed	17	Centaur	No		
23	3-Apr-09	LC-41	Launched					
24	17-Jun-09	LC-41	Scrubbed	1	Range Conflict	No		
25	18-Jun-09	LC-41	Launched			Yes	92	Weather
26	8-Sep-09	LC-41	Launched					
27	18-Oct-09	VAFB SLC-3E	Launched					
28	14-Nov-09	LC-41	Scrubbed	9	FTS	No		
29	23-Nov-09	LC-41	Launched				65	Winds Aloft
30	10-Feb-10	LC-41	Scrubbed	1	Weather	No		
31	11-Feb-10	LC-41	Launched					
32	22-Apr-10	LC-41	Launched					
33	14-Aug-10	LC-41	Launched					
34	21-Sep-10	VAFB SLC-3E	Launched				30	COLA; Range Intrusion
35	5-Mar-11	LC-41	Launched			Yes	97	
36	14-Apr-11	VAFB SLC-3E	Launched					
37	6-May-11	LC-41	Scrubbed	1	Weather	No		
38	7-May-11	LC-41	Launched					
39	5-Aug-11	LC-41	Launched			Yes	51	Range Intrusion; Ground Systems
40	26-Nov-11	LC-41	Launched					
Continued								

Atlas V Launch Attempt History (Continued)								
#	Attempt Date	Launch Site	Outcome	Scrub Duration	Scrub Category	T-0 Delay?	T-0 Delays (minutes)	T-0 Delay Reason
41	16-Feb-12	LC-41	Scrubbed	1	Winds Aloft	No		
42	17-Feb-12	LC-41	Scrubbed	7	Winds Aloft & Range Weather	No		
43	24-Feb-12	LC-41	Launched					
44	3-May-12	LC-41	Scrubbed	1	Human Error & Range Systems	No		
45	4-May-12	LC-41	Launched					
46	18-Jun-12	LC-41	Scrubbed	2	Ground Systems	No		
47	20-Jun-12	LC-41	Launched					
48	2-Aug-12	VAFB SLC-3E	Scrubbed	42	Range Systems	No		
49	23-Aug-12	LC-41	Scrubbed	1	Launch Vehicle	No		
50	24-Aug-12	LC-41	Scrubbed	1	Range Systems	No		
51	25-Aug-12	LC-41	Scrubbed	5	Weather	No		
52	30-Aug-12	LC-41	Launched					
53	13-Sep-12	VAFB SLC-3E	Launched					
54	11-Dec-12	LC-41	Launched					
55	30-Jan-13	LC-41	Launched					
56	11-Feb-13	VAFB SLC-3E	Launched					
57	19-Mar-13	LC-41	Launched					
58	15-May-13	LC-41	Launched					
59	19-Jul-13	LC-41	Launched			Yes	12	Winds Aloft
60	18-Sep-13	LC-41	Launched					
61	18-Nov-13	LC-41	Launched					
62	6-Dec-13	VAFB SLC-3E	Launched				1.5	Unspecified
63	24-Jan-14	LC-41	Launched			Yes	28	Spacecraft
64	3-Apr-14	VAFB SLC-3E	Launched					
65	10-Apr-14	LC-41	Launched					
66	22-May-14	LC-41	Launched			Yes	4	Unspecified
67	1-Aug-14	LC-41	Launched					
68	13-Aug-14	VAFB SLC-3E	Launched					
69	16-Sep-14	LC-41	Launched			Yes	146	Weather
70	29-Oct-14	LC-41	Launched					
71	11-Dec-14	VAFB SLC-3E	Scrubbed	2	Weather	No		
72	13-Dec-14	VAFB SLC-3E	Launched			Yes	6	Unspecified
73	22-Jan-15	LC-41	Launched			Yes	21	Winds Aloft; Range Systems
74	12-Mar-15	LC-41	Launched					

Falcon 9 Launch Attempt History								
#	Attempt Date	Launch Site	Outcome	Scrub Duration	Scrub Category	T-0 Delay?	T-0 Delays (minutes)	T-0 Delay Reason
1	4-Jun-10	LC-40	Launched			Yes	240	Flight Termination System; Range Intrusion; Engine shutdown
2	7-Dec-10	LC-40	Scrubbed	1	Upper Stage Engine			
3	8-Dec-10	LC-40	Launched			Yes	202	Ground systems
4	19-May-12	LC-40	Scrubbed	3	Engine Start			
5	22-May-12	LC-40	Launched					
6	8-Oct-12	LC-40	Launched					
7	1-Mar-13	LC-40	Launched					
8	29-Sep-13	VAFB SLC-4E	Launched					
9	25-Nov-13	LC-40	Scrubbed	3	Core Stage MPS			
10	28-Nov-13	LC-40	Scrubbed	3	Engine Start			
11	1-Dec-13	LC-40	Scrubbed	1	Range Conflict			
12	2-Dec-13	LC-40	Scrubbed	1	Core Stage MPS			
13	3-Dec-13	LC-40	Launched					
14	3-Jan-14	LC-40	Scrubbed	3	Launch Vehicle			
15	6-Jan-14	LC-40	Launched					
16	14-Apr-14	LC-40	Scrubbed	4	Core Stage MPS			
17	18-Apr-14	LC-40	Launched					
18	10-May-14	LC-40	Scrubbed	33	Static Test Problem			
19	12-Jun-14	LC-40	Scrubbed	8	Spacecraft			
20	20-Jun-14	LC-40	Scrubbed	1	Upper Stage MPS			
21	21-Jun-14	LC-40	Scrubbed	1	Launch Weather			
22	22-Jun-14	LC-40	Scrubbed	22	Launch Vehicle			
23	14-Jul-14	LC-40	Launched			Yes	153	Unspecified issues
24	5-Aug-14	LC-40	Launched			Yes	166	Engine Start Abort
25	26-Aug-14	LC-40	Scrubbed	12	Test Mishap			
26	7-Sep-14	LC-40	Launched			Yes	194	Weather
27	20-Sep-14	LC-40	Scrubbed	1	Weather: Range & Launch			
28	21-Sep-14	LC-40	Launched					
29	19-Dec-14	LC-40	Scrubbed	18	Static Test Problem			
30	6-Jan-15	LC-40	Scrubbed	4	Upper Stage MPS			
31	10-Jan-15	LC-40	Launched					
32	8-Feb-15	LC-40	Scrubbed	1	Range Systems			
33	9-Feb-15	LC-40	Scrubbed	1	Launch Weather			
34	10-Feb-15	LC-40	Scrubbed	1	Winds Aloft			
35	11-Feb-15	LC-40	Launched					
36	1-Mar-15	LC-40	Launched					

Delta IV Launch Attempt History								
Attempt	Planned Date	Launch Site	Outcome	Scrub Duration	Scrub Category	T-0 Delay?	T-0 Delays (minutes)	T-0 Delay Reason
1	16-Nov-02	LC-37	Scrubbed	3	Upper Stage MPS: Launch Weather			
2	19-Nov-02	LC-37	Scrubbed	1	Core Stage MPS plus: Launch Weather; Cores Stage Avionics; Ground Systems			
3	20-Nov-02	LC-37	Launched					
4	8-Mar-03	LC-37	Scrubbed	1	Ground Systems			
5	9-Mar-03	LC-37	Scrubbed	2	Winds Aloft			
6	11-Mar-03	LC-37	Launched			Yes	78	Unspecified
7	4-Aug-03	LC-37	Scrubbed	24				
8	28-Aug-03	LC-37	Scrubbed	1	Launch Weather			
9	29-Aug-03	LC-37	Launched					
10	10-Dec-04	LC-37	Scrubbed	1	Launch Weather			
11	11-Dec-04	LC-37	Scrubbed	1	Ground Systems			
12	12-Dec-04	LC-37	Scrubbed	9	Ground Systems			
13	21-Dec-04	LC-37	Launched			Yes	134	Unspecified
14	15-Aug-05	LC-37	Scrubbed	1	Upper Stage MPS			
15	16-Aug-05	LC-37	Scrubbed	282	Launch Vehicle Batteries (40 Days) Extende delay due to GOES-N			
16	25-May-06	LC-37	Launched					
17	28-Jun-06	VAFB SLC-6	Launched			Yes	19	Launch Weather
18	4-Nov-06	VAFB SLC-6	Launched			Yes	1	COLA
19	11-Nov-07	LC-37	Launched			Yes	11	Instrumentation link
20	13-Jan-09	LC-37	Scrubbed	1	Launch Vehicle Processing			
21	14-Jan-09	LC-37	Scrubbed	2	Ground Systems			
22	15-Jan-09	LC-37	Scrubbed	1	Launch Weather			
23	17-Jan-09	LC-37	Launched			Yes	134	Undisclosed
24	26-Jun-09	LC-37	Scrubbed	1	Launch Weather			
25	27-Jun-09	LC-37	Launched			Yes	37	Launch Weather
26	2-Dec-09	LC-37	Scrubbed	1	Weather			
27	3-Dec-09	LC-37	Scrubbed	2	Winds Aloft (25 minutes) Range Weather (41 minutes) Ground Systems (data card failure)			
28	6-Dec-09	LC-37	Launched			Yes	82	Winds Aloft
Continued								

Delta IV Launch Attempt History (Continued)								
Attempt	Planned Date	Launch Site	Outcome	Scrub Duration	Scrub Category	T-0 Delay?	T-0 Delays (minutes)	T-0 Delay Reason
29	2-Mar-10	LC-37	Scrubbed	1	Launch Vehicle (TVC)			
30	3-Mar-10	LC-37	Scrubbed	1	Ground Systems			
31	4-Mar-10	LC-37	Launched			Yes	55	Ground Systems; Winds Aloft
32	20-May-10	LC-37	Scrubbed	1	Ground Systems			
33	21-May-10	LC-37	Scrubbed	2	Payload			
34	23-May-10	LC-37	Scrubbed	1				
35	24-May-10	LC-37	Scrubbed	4	Main Engine Start			Main Engine Start
36	27-May-10	LC-37	Launched					
37	18-Nov-10	LC-37	Scrubbed	2	Ground Systems			
38	19-Nov-10	LC-37	Scrubbed	1	Side Booster MPS			
39	21-Nov-10	LC-37	Launched					
40	20-Jan-11	VAFB SLC-6	Launched					
41	11-Mar-11	LC-37	Launched			Yes	41	Winds Aloft
42	14-Jul-11	LC-37	Scrubbed	1	Payload			
43	15-Jul-11	LC-37	Scrubbed	1	Weather			
44	16-Jul-11	LC-37	Launched					
45	20-Jan-12	LC-37	Launched					
46	3-Apr-12	VAFB SLC-6	Launched					
47	28-Jun-12	LC-37	Scrubbed	1	Hurricane			
48	29-Jun-12	LC-37	Launched			Yes	106	Upper Stage MPS; Core Stage MPS
49	4-Oct-12	LC-37	Launched					
50	24-May-13	LC-37	Scrubbed	1	Ground Systems			
51	25-May-13	LC-37	Launched					
52	8-Aug-13	LC-37	Launched					
53	28-Aug-13	VAFB SLC-6	Launched					
54	21-Feb-14	LC-37	Launched					
55	15-May-14	LC-37	Scrubbed	1	Weather			
56	16-May-14	LC-37	Launched					
57	23-Jul-14	LC-37	Scrubbed	1	Ground Systems			
58	24-Jul-14	LC-37	Scrubbed	1	Launch Weather			
59	25-Jul-14	LC-37	Scrubbed	1				
60	26-Jul-14	LC-37	Scrubbed	2				
61	28-Jul-14	LC-37	Launched					
62	4-Dec-14	LC-37	Scrubbed	1	Range Intrusion; Propellant Conditioning; Launch Weather (surface winds); Core Stage MPS (fill & drain valve)			
63	5-Dec-14	LC-37	Launched					
64	25-Mar-15	LC-37	Launched					

ACKNOWLEDGEMENTS

We would like to thank Jeff Hanley, the Principal Director of the Human Exploration and Spaceflight division, and John Chobany, the Director of the Space Architecture Department, at The Aerospace Corporation for supporting this effort. We very much appreciated the comments and suggestions provided by Michael Moore, Robert Minnichelli, and Gary Lin.

REFERENCES

- [1] Spaceflight Now Web Site:
<http://spaceflightnow.com>
- [2] Cleary, M.C., "Evolved Expendable Launch Operations at Cape Canaveral, 2002-2009," 45th Space Wing History Office.

- [3] Cates, G. (2014, March). Space shuttle launch probability analysis: Understanding history so we can predict the future. In Aerospace Conference, 2014 IEEE (pp. 1-17). IEEE.
- [4] Chang, I-Shih. "Investigation of space launch vehicle catastrophic failures." *Journal of spacecraft and rockets* 33.2 (1996): 198-205.
- [5] Altino, K. M., Burns, K. L., Barbré Jr, R. E., & Leahy, F. B. Analyzing the Impacts of Natural Environments on Launch and Landing Availability for NASA's Exploration Systems Development Programs. In SpaceOps 2014 13th International Conference on Space Operations, AIAA, Pasadena, CA
- [6] Harrington, B.E., Picka, B.A., and Cordova, H.S., "Space Shuttle Day-of-Launch Trajectory Design Operations," AIAA 2011-7197
- [7] National Hurricane Center, Tropical Cyclone Climatology Web site:
<http://www.nhc.noaa.gov/climo/#bac>
- [8] Morisette, J.T. and Khorram, S., "Exact Binomial Confidence Interval for Proportions," Photogrammetric Engineering & Remote Sensing, April 1998, pages 281-283.
- [9] Space Shuttle Missions Summary, NASA/TM-2011-216142.
- [10] Johnston, Smith L. "Flight Crew Health Stabilization Program," JSC-22538 Rev. D (2010)
- [11] Barbré Jr, R. E., & Keller, V. W. P2. 2 SEA STATE AND WEATHER ASSESSMENT CAPABILITY FOR NASA'S CONSTELLATION PROGRAM,
<http://ntrs.nasa.gov/search.jsp?R=20080013554>
- [12] Rockwell Automation's Arena Web site:
<http://www.arenasimulation.com/>
- [13] Banks, J., J. S. Carson, and B. L. Nelson. DM Nicol, Discrete-Event System Simulation. Englewood Cliffs, NJ, USA: Prentice hall, 2000.
- [14] Law, A. M. "Simulation modeling and analysis." McGraw-Hill series in industrial engineering and management science Show all parts in this series (2007).

BIOGRAPHY



Grant Cates is a Senior Project Leader at The Aerospace Corporation since February 2014. Prior to that he was a Chief Scientist at SAIC. He retired from NASA in 2006 after 25 combined years in federal service, including 7 years on active duty in the Air Force. At NASA he served in varying capacities on the Space Shuttle Program, including Space Shuttle Columbia Vehicle Manager and Flow Director. He received a Ph.D. in Industrial Engineering from the University of Central Florida in 2004.



Kara Schmitt is an Engineering Specialist in the Space Architecture Department at The Aerospace Corporation since August 2014. She has worked in the domains of Spaceflight, and Nuclear Power to improve safety in ways that are practical, simple, and implementable. Prior to joining Aerospace she was Vice President of Research and Development at Micro Aerospace Solutions, Inc. She received a Ph.D. in Human Centered Design from the Florida Institute of Technology in 2013